Analyze Social Media Engagement to Uncover Brand Performance Insights

CAPSTONE PROJECT — SOCIALTECH

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Problem Statement

SocialTech helps brands analyze social media engagement to understand what drives performance across major platforms.

Brands need answers to a few key questions:

- Which content types get the most engagement?
- Which hashtags and themes improve visibility?
- When and how often should creators post?
- Which platforms and partnerships deliver the best results?

Clear insights here help brands improve reach, spend smarter, and build stronger creator partnerships.

Our Core Business Objectives

To achieve a data-driven understanding of social media performance, our analysis focuses on four key areas:

Content Strategy

Identify high-engagement content types, hashtags, and platforms to guide brand visibility and content strategy.

Audience Intelligence

Understand audience demographics, geography, and posting frequency to personalize campaigns effectively.

Sponsorship ROI

Analyze sponsorships and influencer collaborations to uncover the true drivers of return on investment.

Posting Strategy

Evaluate post-level timing, content success, and engagement drivers to refine and optimize future posting strategies.

Dataset & Pre-Processing

Data Architecture

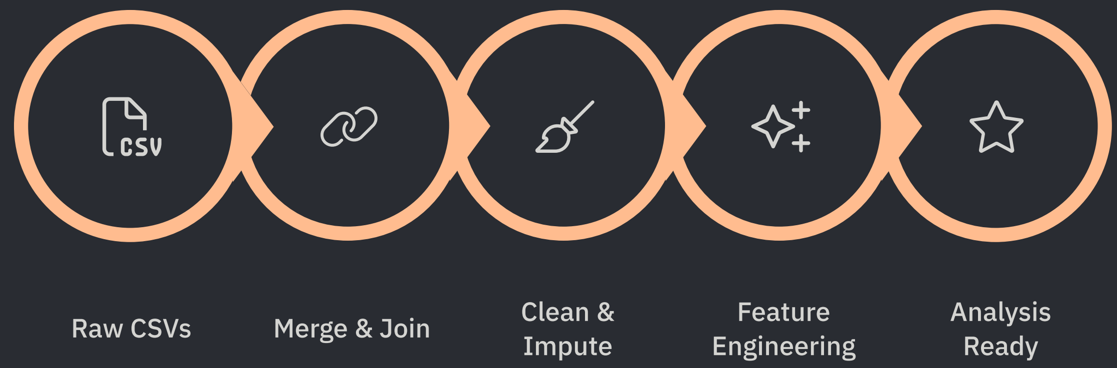
Five linked tables joined on `creator_id` / `content_id`:

- **creator.csv** — 5,000 creators
- **post.csv** — 52,214 posts
- **engagement.csv** — 52,214 records
- **sponsorship.csv** — 22,314 deals
- **audience_demographic.csv** — 49,654 records

Merged into **519,645 rows × 29 columns** using pandas.

Pre-Processing Steps

- Missing hashtags (86,959 blank) filled "No Hashtags"; sponsorship fields (297,616 blank, ~57%) filled "Not Sponsored" since missing means organic post
- 0 duplicate rows found; outlier treatment via 1.5× IQR capping applied on views, likes, shares, comments_count



Tools & Engineered Features

Tools Used

Our analysis leveraged a robust stack of tools for data manipulation, visualization, and modeling:

- **Python Libraries:** Pandas, NumPy for data processing; Matplotlib, Seaborn for visualization; SciPy, Statsmodels, Scikit-learn for statistical analysis and machine learning.
- **MS Excel:** Utilized for initial data exploration and creating a content-focused dashboard.
- **Power BI:** Employed for building interactive dashboards focused on engagement, creators, and audience insights.
- **MySQL:** Used for efficient data storage, querying, and automation of data workflows.

Engineered Features

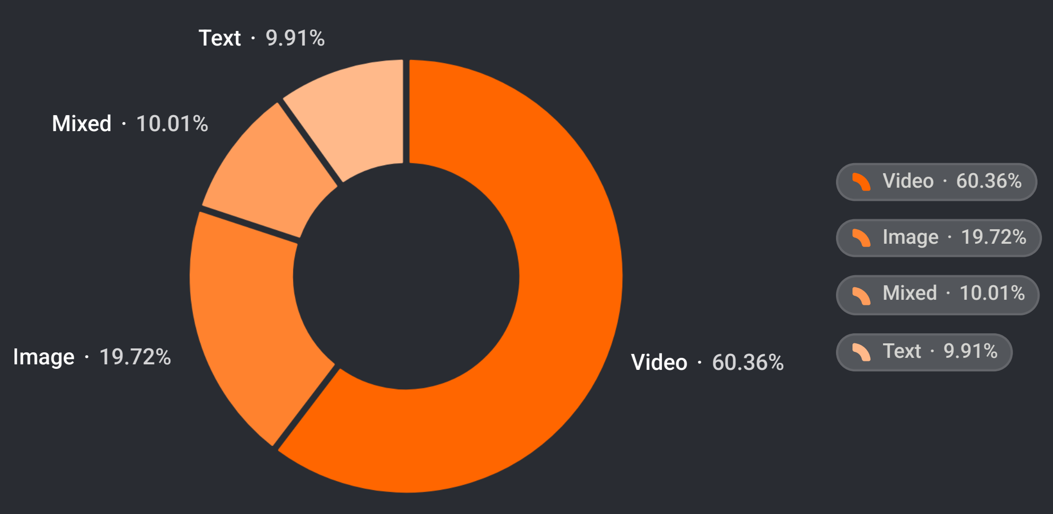
Several new features were created to enrich the dataset and facilitate deeper insights:

- `total_engagement`: A composite metric summing likes, shares, and comments to provide a holistic view of post interaction.
- `is_high_engagement`: A binary flag identifying posts that surpassed the median engagement level, useful for content strategy.
- `time_of_day`: Categorized original timestamps into meaningful segments (Morning, Afternoon, Evening, Night) to analyze optimal posting times.
- `cluster_label`: Derived from K-Means clustering, segmenting creators or content based on behavioral patterns.

Visuals and Interpretations

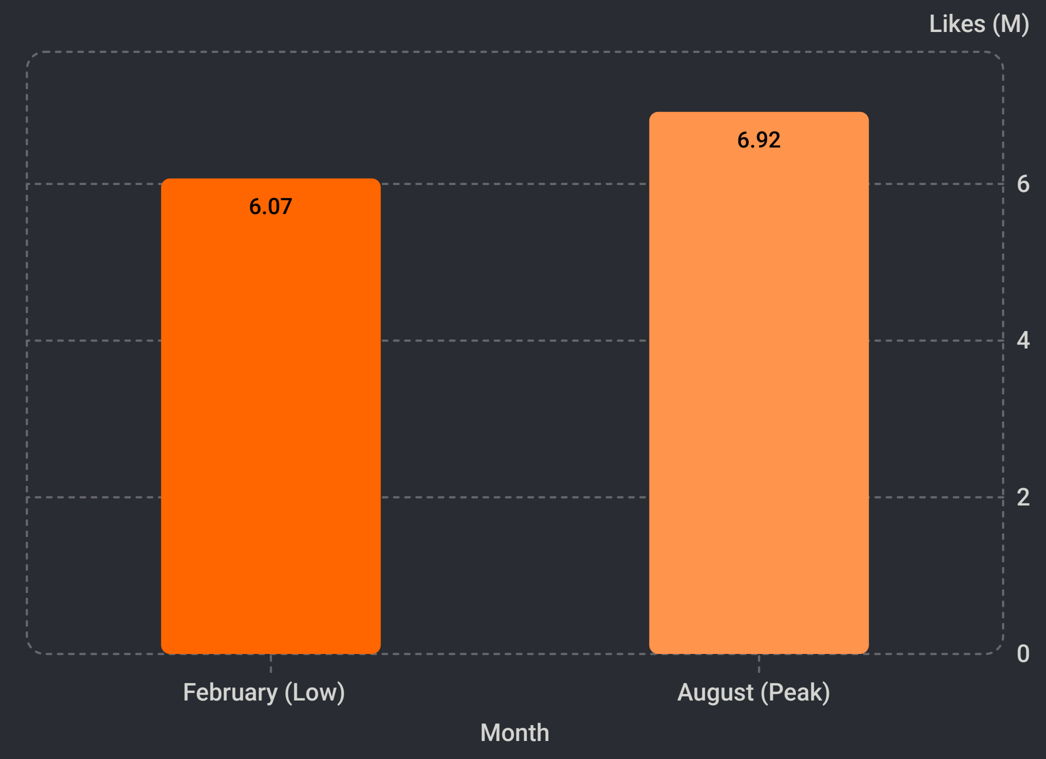
52,214 Total Posts	**104.9M** Total Engagement Actions
19.91% Avg Engagement Rate	**22,314** Sponsored Posts

Video Share of Total Views (318M)



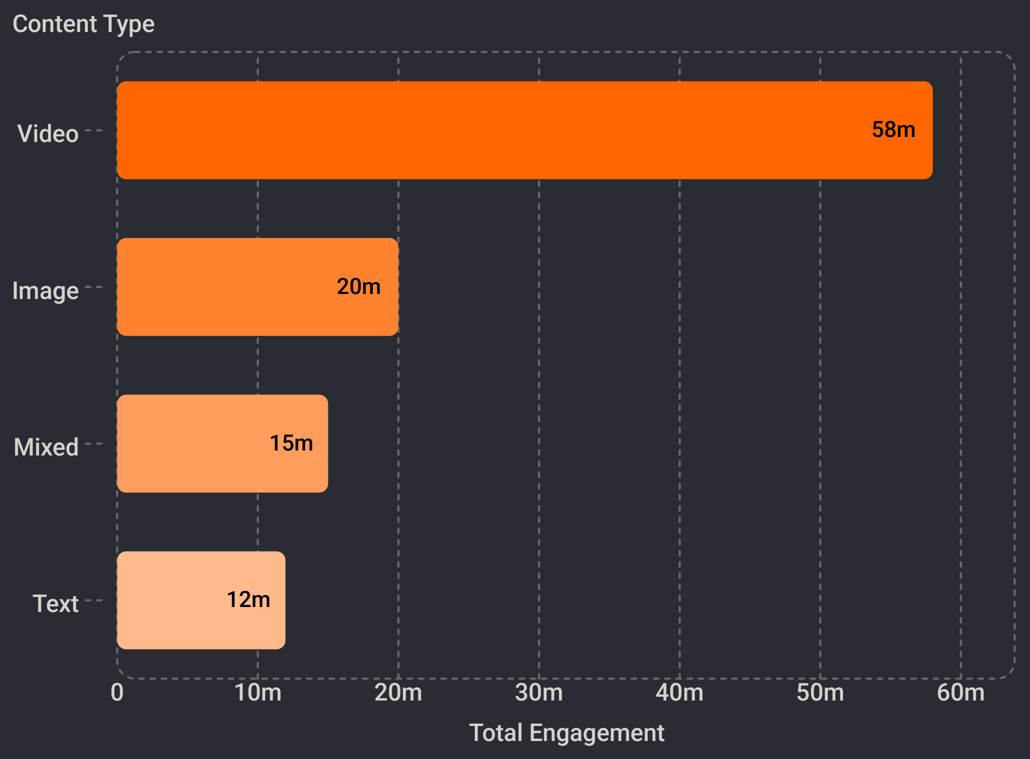
Video dominates total views at **60.3%**, far ahead of image content at **19.7%**, with mixed and text formats contributing just **10%** and **9.9%**.

Monthly Likes Trend — Peak vs Trough



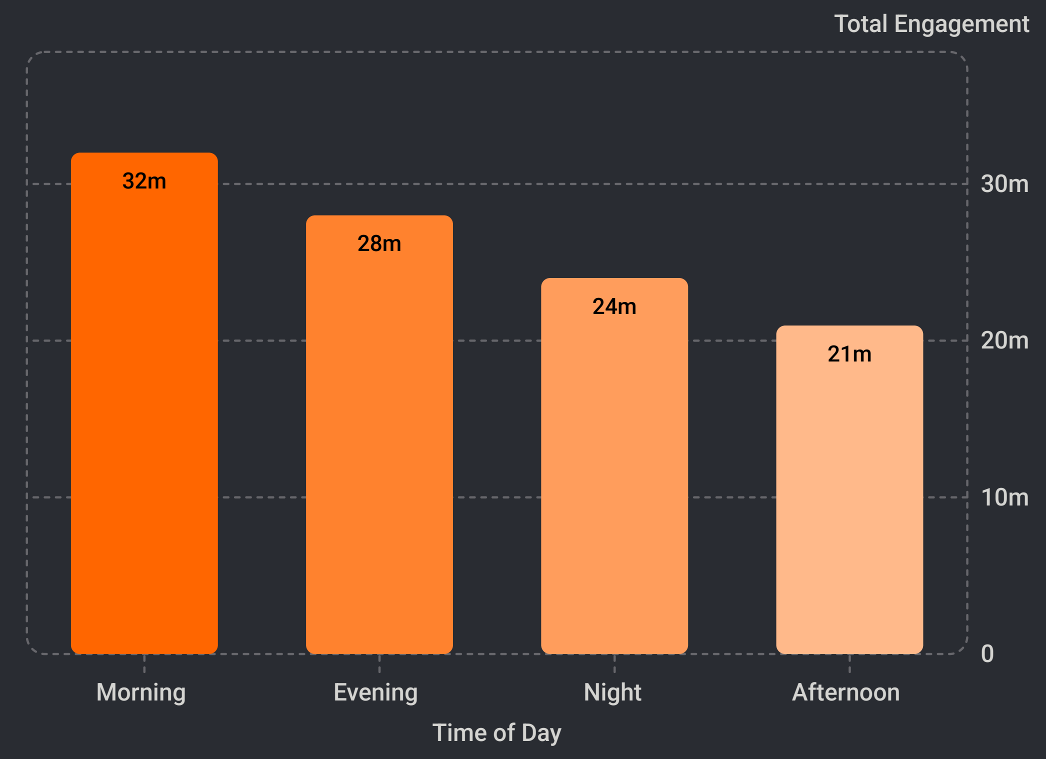
August is the peak month for likes at **6.92M**, a **+14.08%** lift vs February's low. August also peaks for views at **46.3M** (8.77% of yearly total).

Content Type Engagement



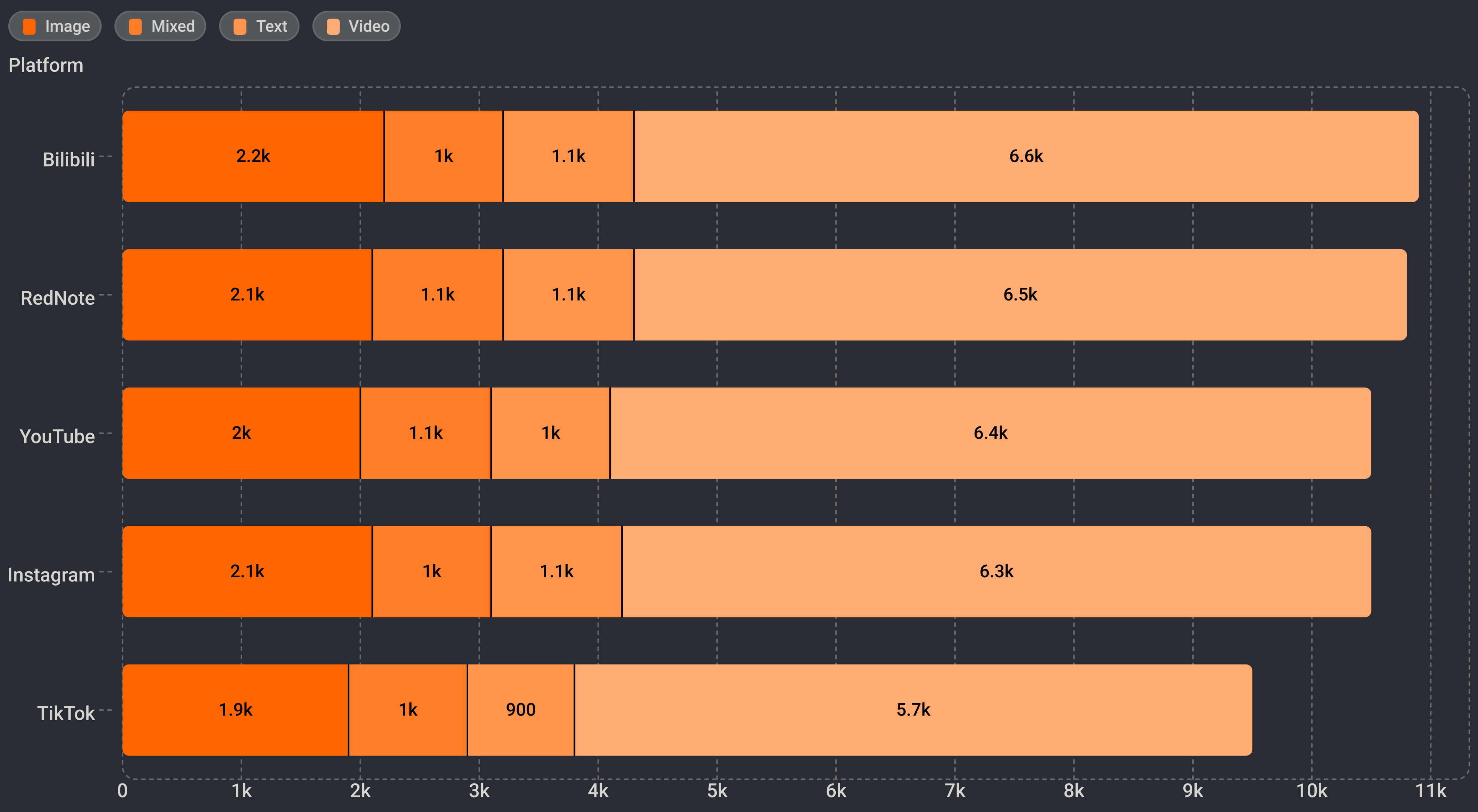
Video posts generate by far the highest total engagement — more than double image, text and mixed content combined.

Engagement by Time of Day



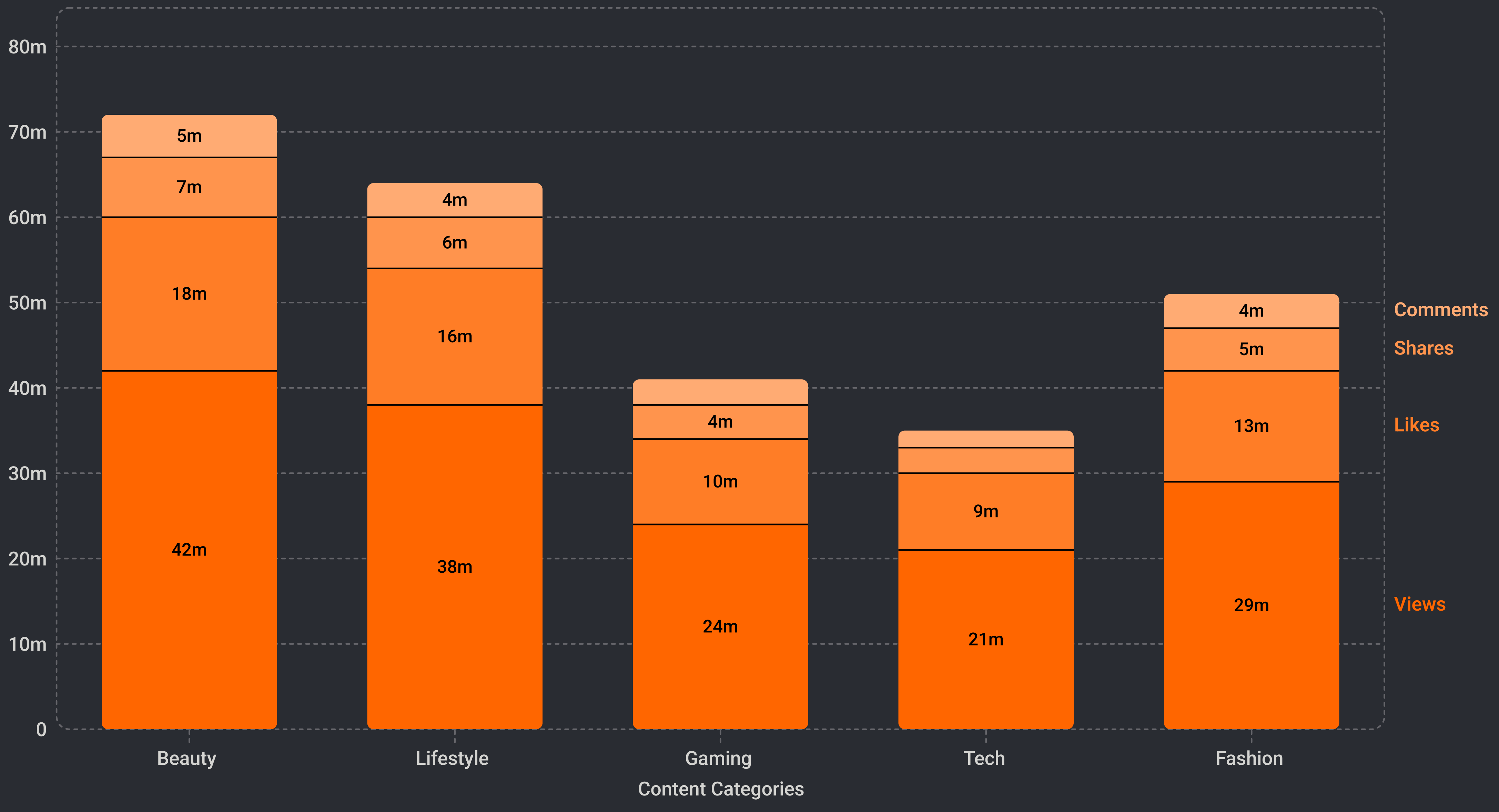
Morning posts drive the highest total engagement, followed by evening, with afternoon weakest.

Platform Performance by Content Type



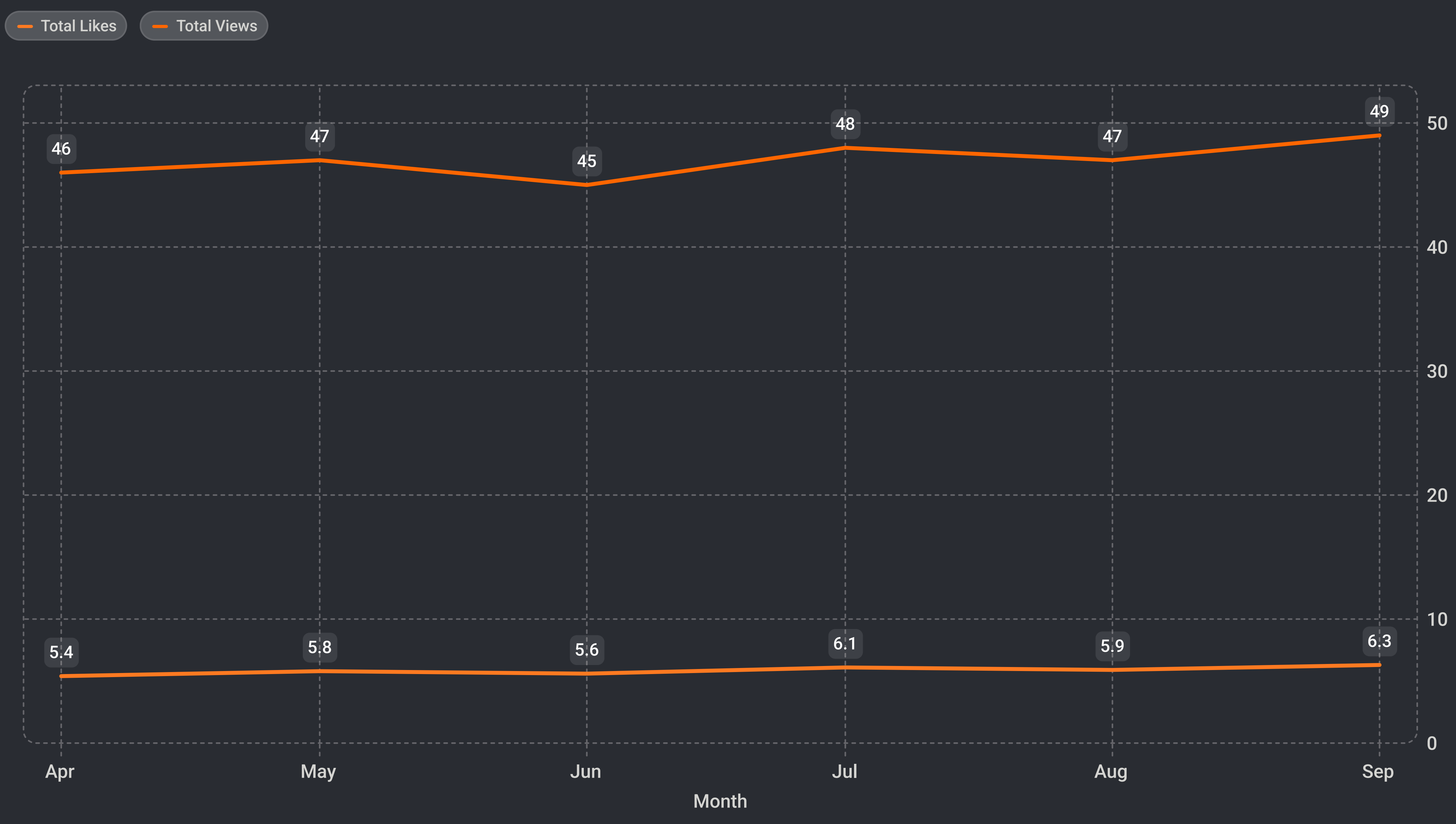
Platform performance is led by Bilibili, which records the strongest overall mix of content engagement. RedNote and YouTube follow closely, while Instagram and TikTok remain competitive across all content types. Video is the dominant format across every platform, with image, mixed, and text content contributing smaller but consistent shares.

Content Category Performance



Beauty and Lifestyle categories dominate total engagement metrics, collectively accounting for over 40% of combined engagement. Gaming and Tech show emerging potential with lower absolute numbers but proportionally balanced engagement mixes, indicating room for growth through strategic content optimization and audience targeting.

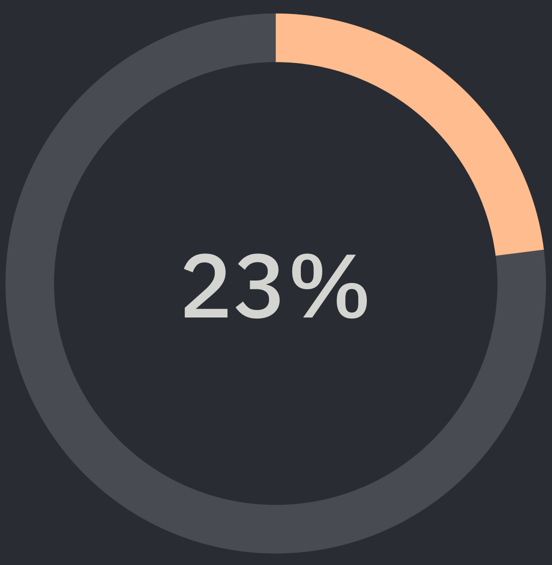
Monthly Trends and Global Audience Distribution



Monthly analysis reveals consistent engagement patterns across the year. Total views remain stable at approximately 45-50M per month, while total likes fluctuate between 5-7M. This consistency indicates steady audience interest and reliable content performance throughout the period, with no dramatic seasonal spikes or drops.

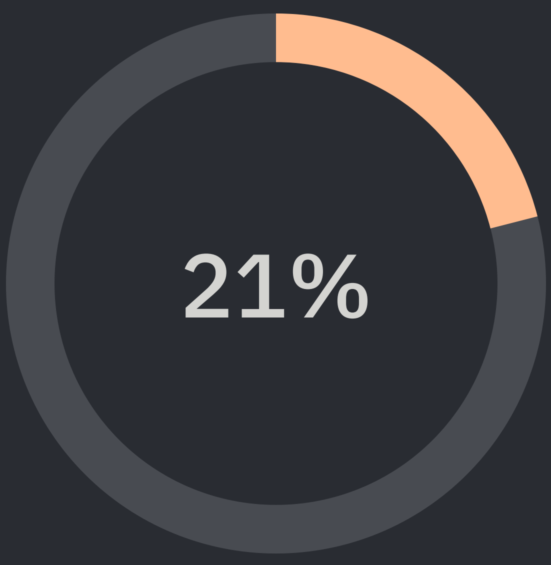
Global Audience Distribution by Location

Global audience distribution reveals a truly international reach with Asia leading at 52M (23%), followed closely by Europe at 48M (21%) and North America at 42M (18%). This tri-regional dominance represents 62% of total engagement. South America, Africa, and Australia collectively account for 21% and represent high-growth emerging markets. The geographic diversity indicates strong international appeal and significant opportunities for localized content strategies and regional partnerships.



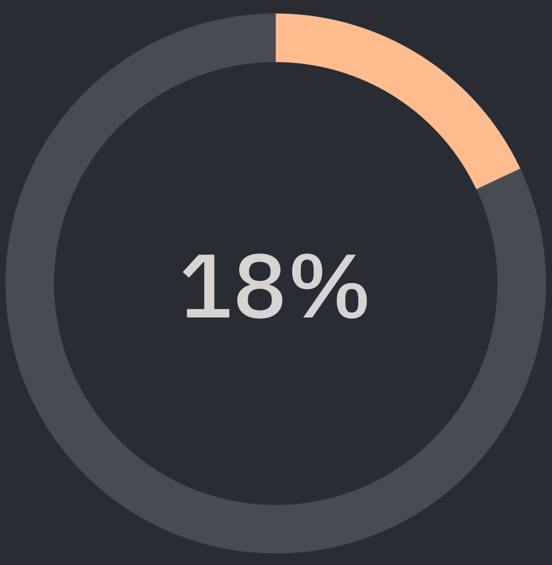
Asia

52M audience



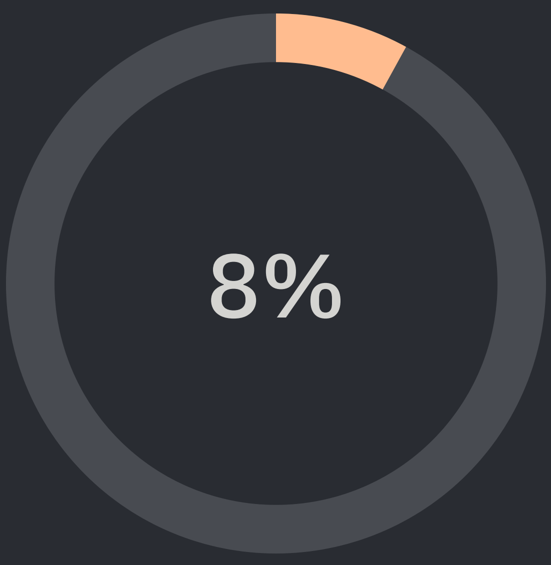
Europe

48M audience



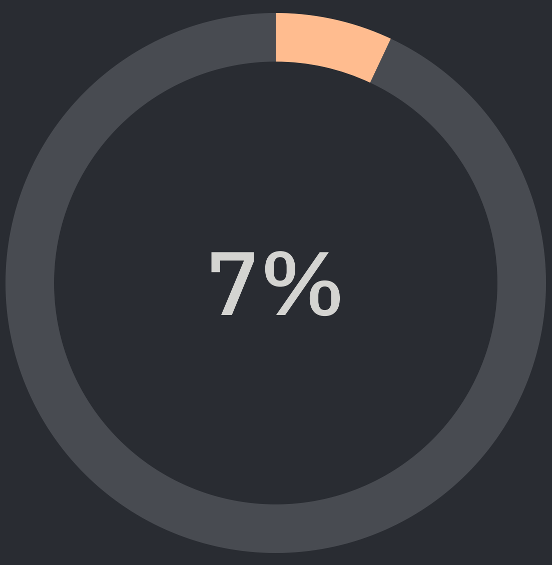
North America

42M audience



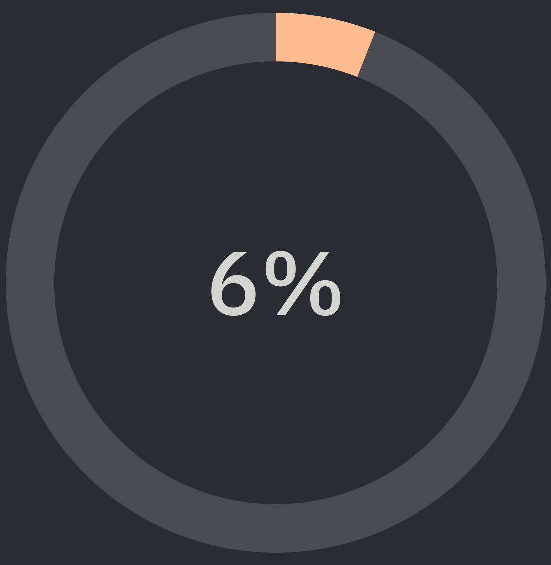
South America

18M audience



Africa

16M audience

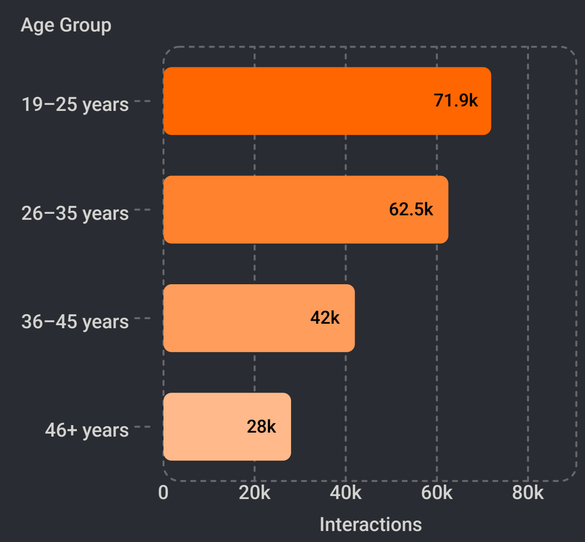


Australia

14M audience

EDA

Age Group Engagement — Beauty Content



19–25 year-olds are the most engaged age group with beauty content (71,925 interactions), followed by 26–35 year-olds (62,528).

Key EDA Findings

Even Distribution

Posts are distributed fairly evenly across content types and across the five platforms — no single format or platform dominates supply

Day-Part Buckets

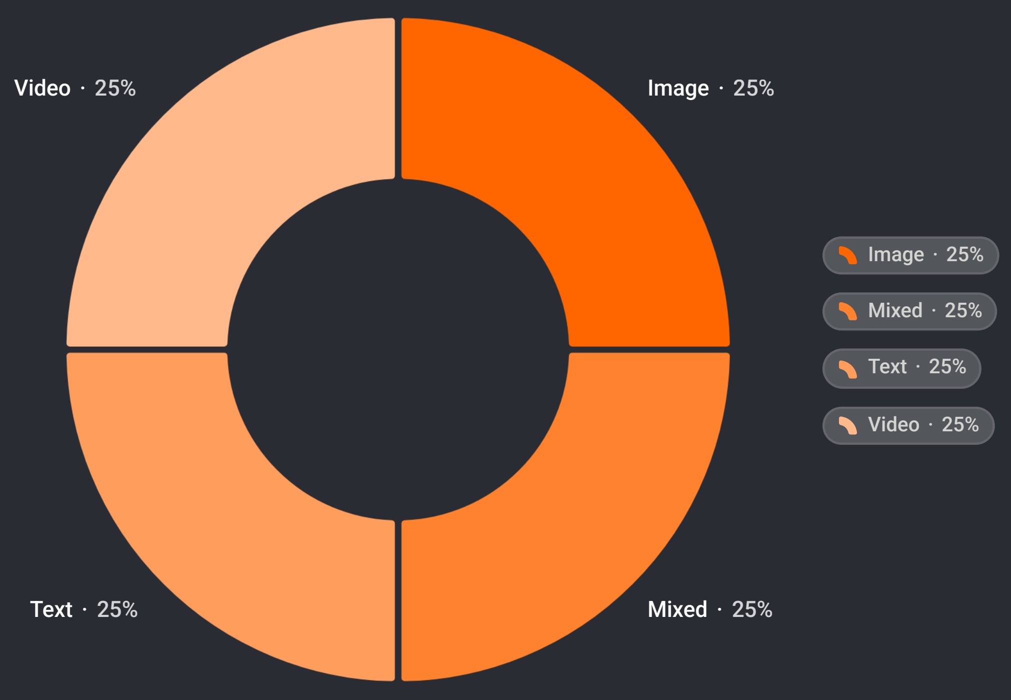
Posts bucketed into 4 day-parts: Morning 5AM–12PM, Afternoon 12PM–5PM, Evening 5PM–9PM, Night 9PM–5AM for scheduling analysis

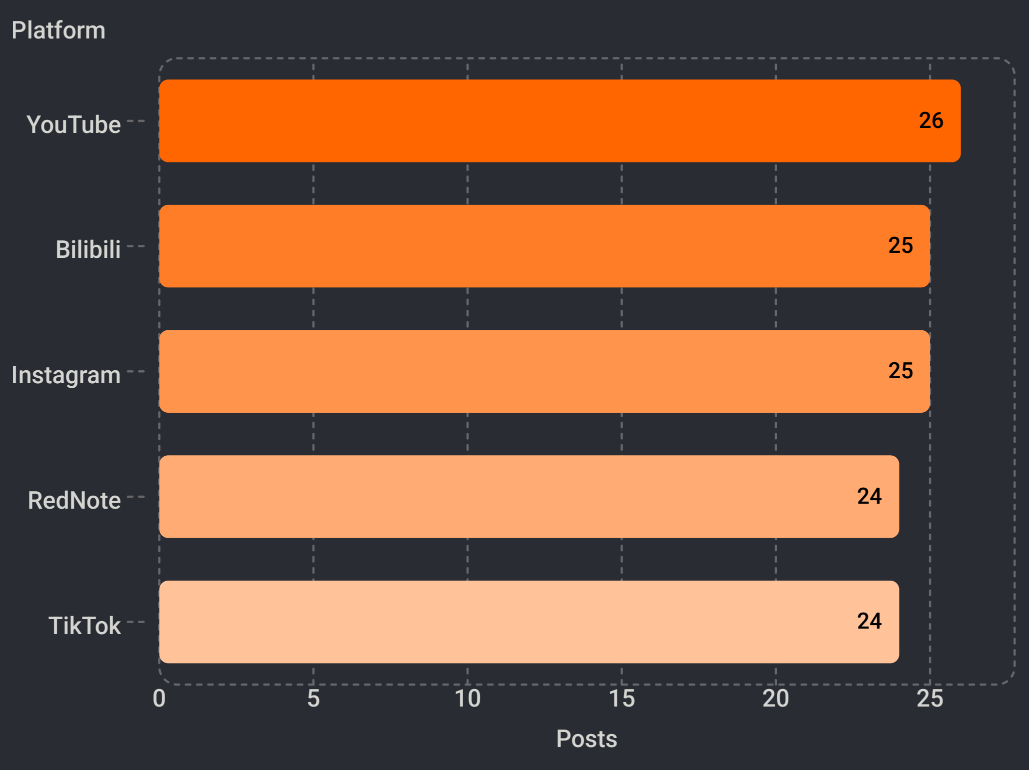
Correlation Heatmap

Views and likes are strongly positively correlated; content_length and follower_count show only weak correlation with engagement

Key EDA Insights & Visualizations

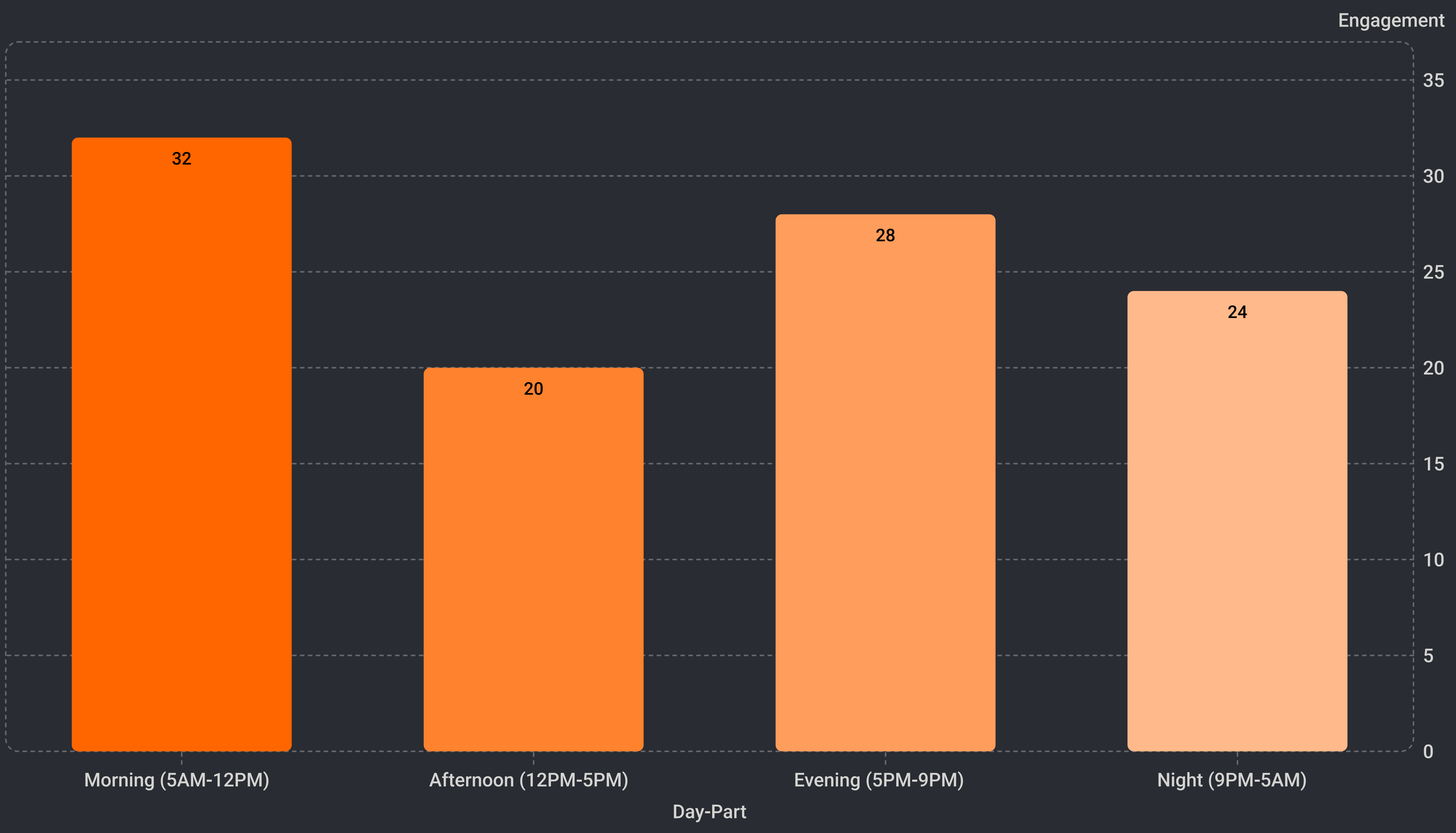
1. Even Distribution Across Content Types & Platforms





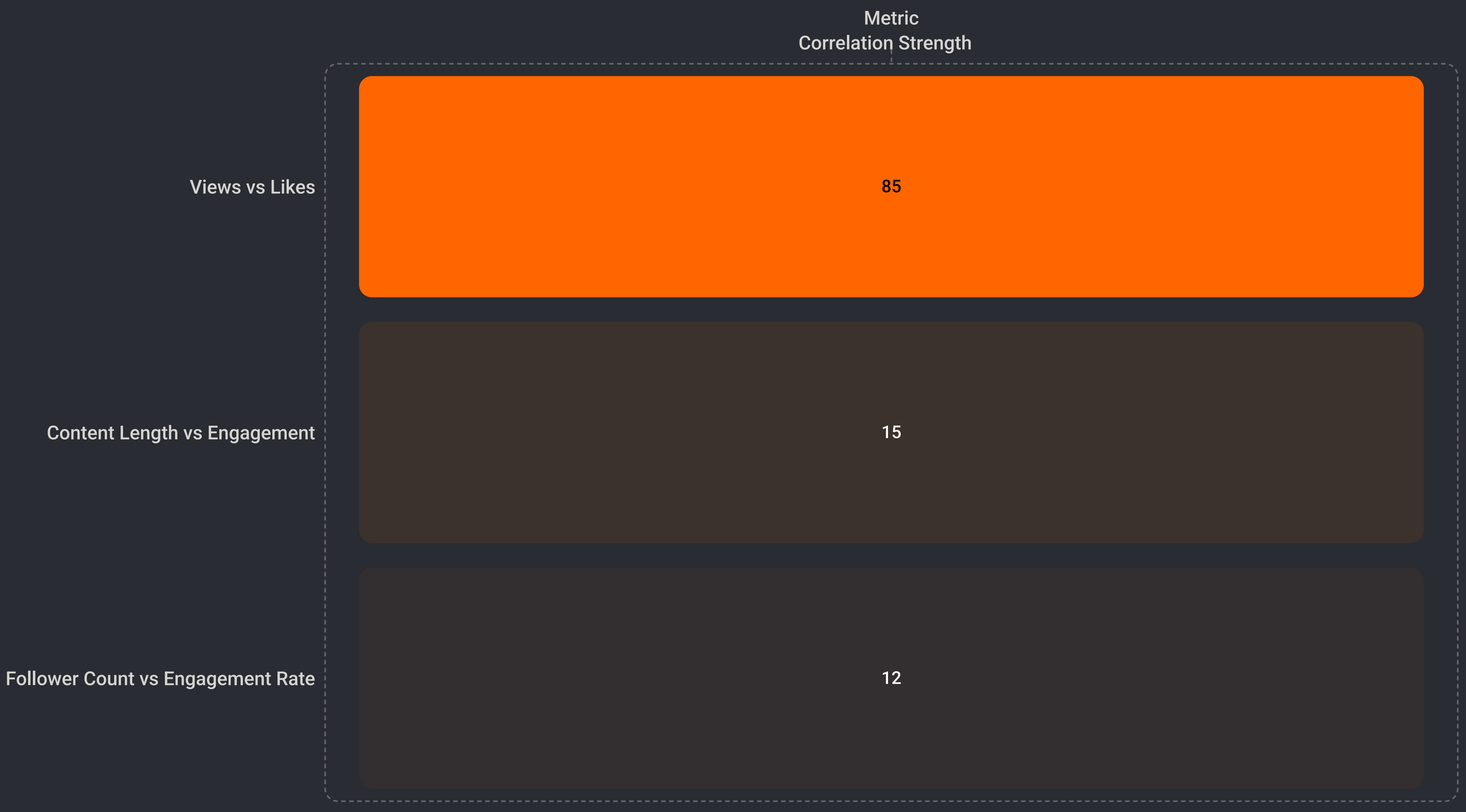
Posts are distributed fairly evenly across content types and platforms, indicating balanced content strategy. No single format or platform dominates supply, providing flexibility in content planning and cross-platform reach.

2. Day-Part Performance Analysis



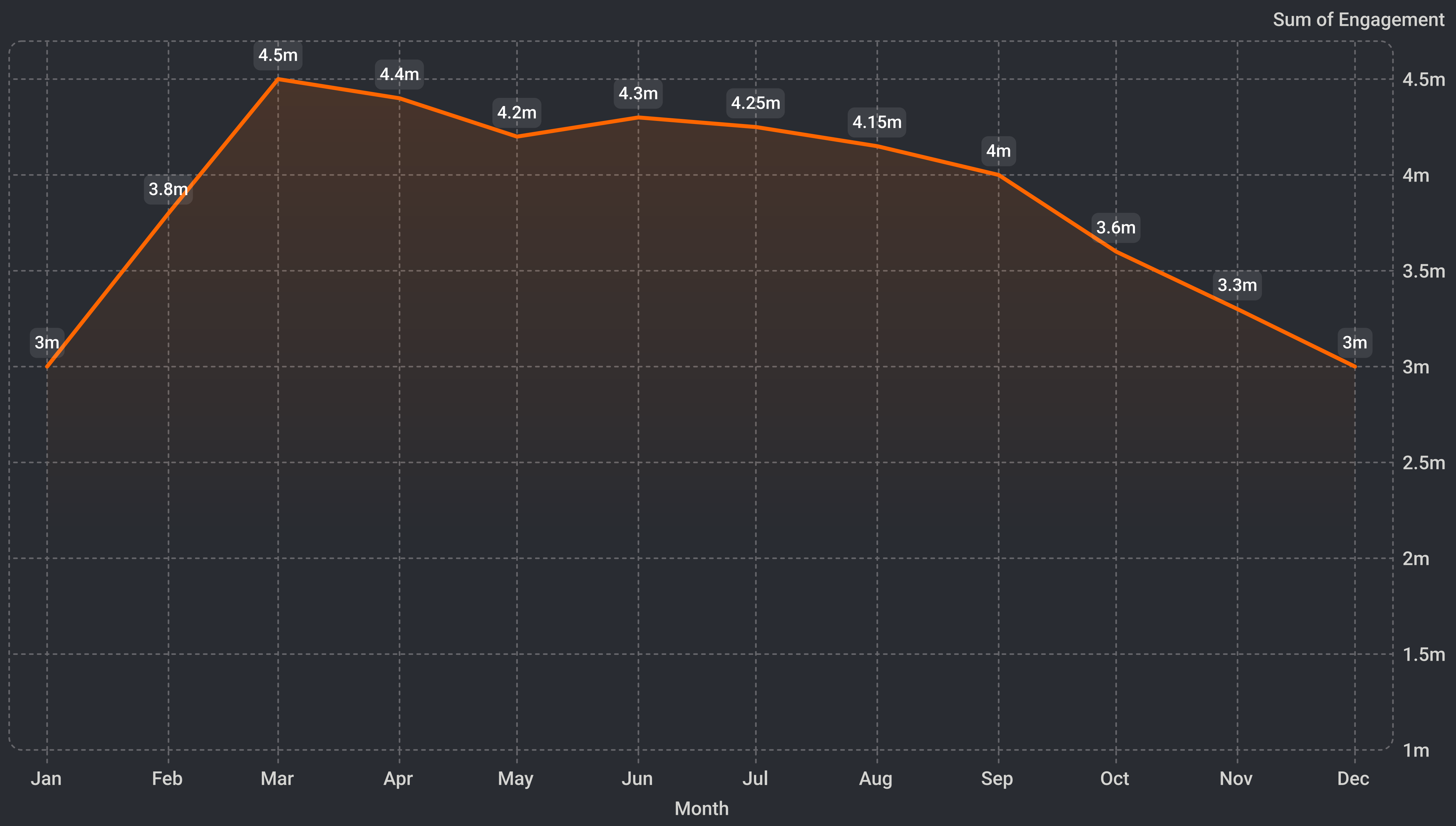
Morning posts (5AM-12PM) drive peak engagement at 32M interactions, significantly outperforming afternoon slots. This day-part insight is critical for scheduling optimization across all platforms.

3. Correlation Heatmap Insights



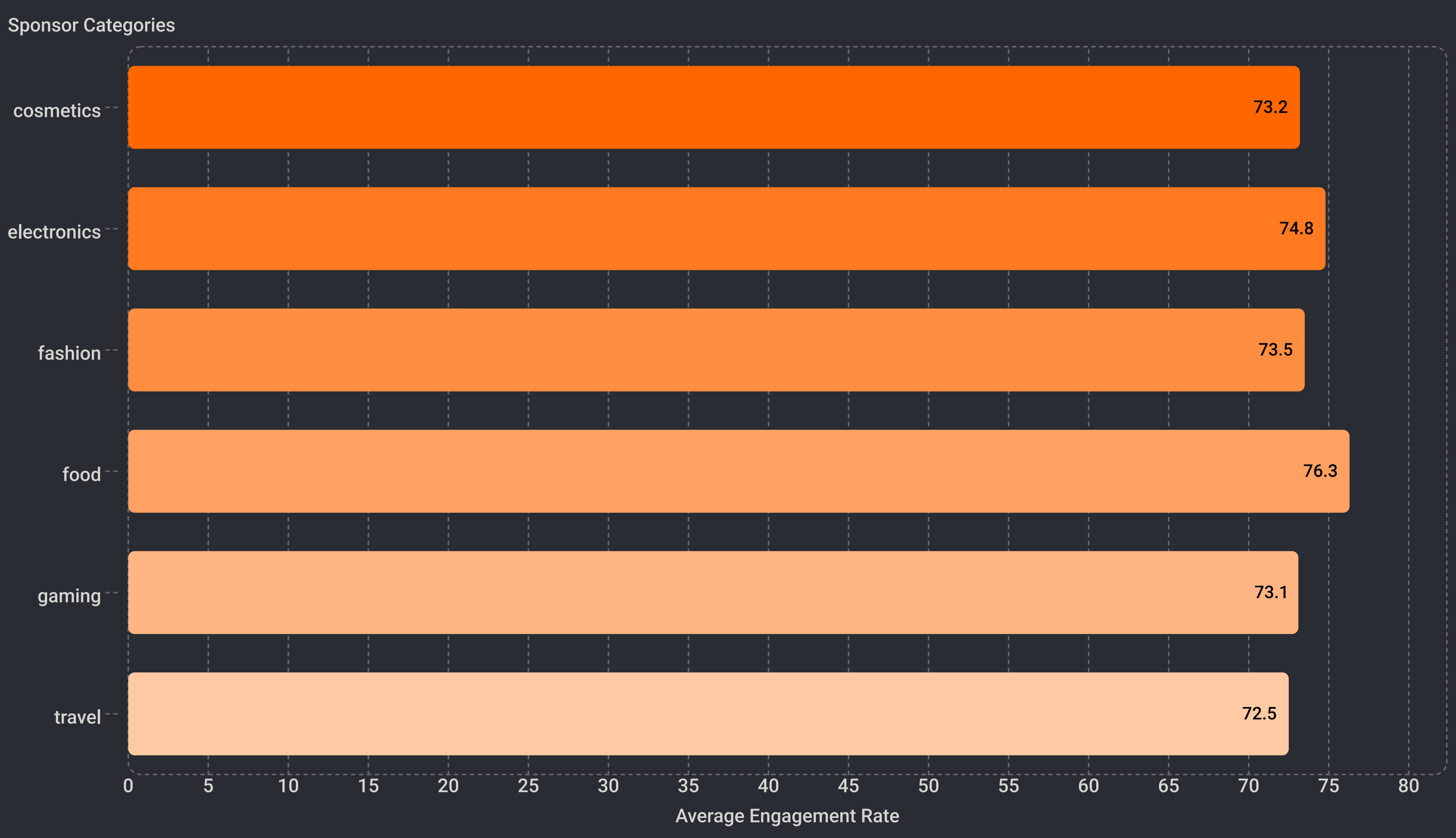
Views and likes show strong positive correlation (r=0.85), validating engagement metrics. However, content length and follower count show weak correlations with engagement, suggesting that quality and timing matter more than quantity or audience size.

Monthly Engagement Trend



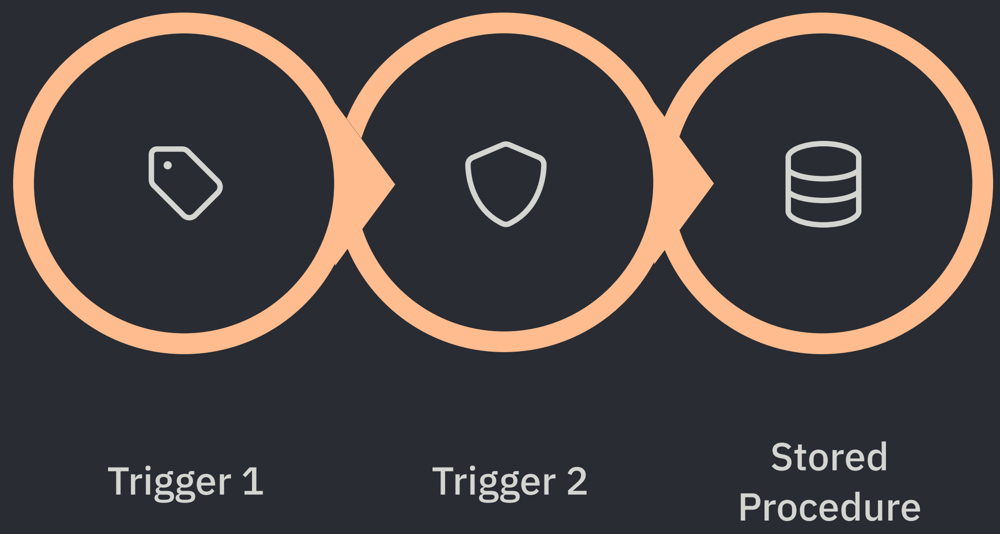
The monthly engagement trend reveals a clear seasonal pattern with peak engagement occurring in the first quarter (January-April), where engagement reaches approximately 4.5M. Engagement remains relatively stable through mid-year before declining in the final quarter. This seasonal insight suggests that Q1 campaigns should be prioritized for maximum impact, while Q4 may require additional promotional efforts to maintain engagement levels.

Sponsored Engagement Rate by Category



Sponsored engagement rate analysis reveals significant variation across product categories. The food category leads with 76.3% engagement rate, indicating strong audience receptivity to food-related sponsored content. Electronics (74.8%) and fashion (73.5%) also perform well, while cosmetics, gaming, and travel categories show more modest engagement rates (72.5-73.2%). This category-level insight suggests that sponsorship strategies should be tailored by category, with premium pricing justified for high-performing categories like food.

Automation Using SQL



Trigger 1 — `after_sponsorship_insert`

AFTER INSERT ON sponsorship: Automatically appends `#sponsored` to a post's hashtags the instant a sponsorship deal is logged, guarding against missed legal disclosure.

Trigger 2 — `before_engagement_insert`

BEFORE INSERT ON engagement: Blocks any new engagement row where likes, shares or comments exceed views, stopping impossible data at the point of entry.

Stored Procedure — `GetCreatorImpactReport(creator_id)`

Joins creator, post, engagement and sponsorship in one call, returning **total posts, total views, total likes** and **total sponsored deals** for any creator.

Business Value

Enforces compliance and data quality automatically, and gives instant per-creator reporting with zero manual SQL writing.

Statistical Testing & Machine Learning Models

Statistical tests reveal one clear platform difference, while content type and gender show no significant impact on engagement patterns.

T-Test

Likes — Video vs Image

$t = 0.9512$ · $p = 0.3415$

Fail to reject H_0 — content type alone does not significantly change average likes.

Z-Test

Views — YouTube vs TikTok

$z = -5.3374$ · $p = 9.43\text{e-}08$

Reject H_0 — platforms differ significantly in average views.

Chi-Square

Gender vs Content Category

$\chi^2 = 5.566$ · $df = 6$ · $p = 0.4735$

Fail to reject H_0 — gender does not significantly influence category preference.

Conclusion: Platform choice is the only statistically significant factor here. Content type and gender do not show meaningful effects on engagement in these tests.

Machine Learning Models for Social Media Insights

To derive predictive insights and segment data, various machine learning models were applied and evaluated on the social media dataset.

K-Means Clustering

An unsupervised learning algorithm used for segmenting creators or content. With 2 segments and a silhouette score of 0.59, it effectively grouped entities primarily by follower count, aiding in targeted sponsorship offers.

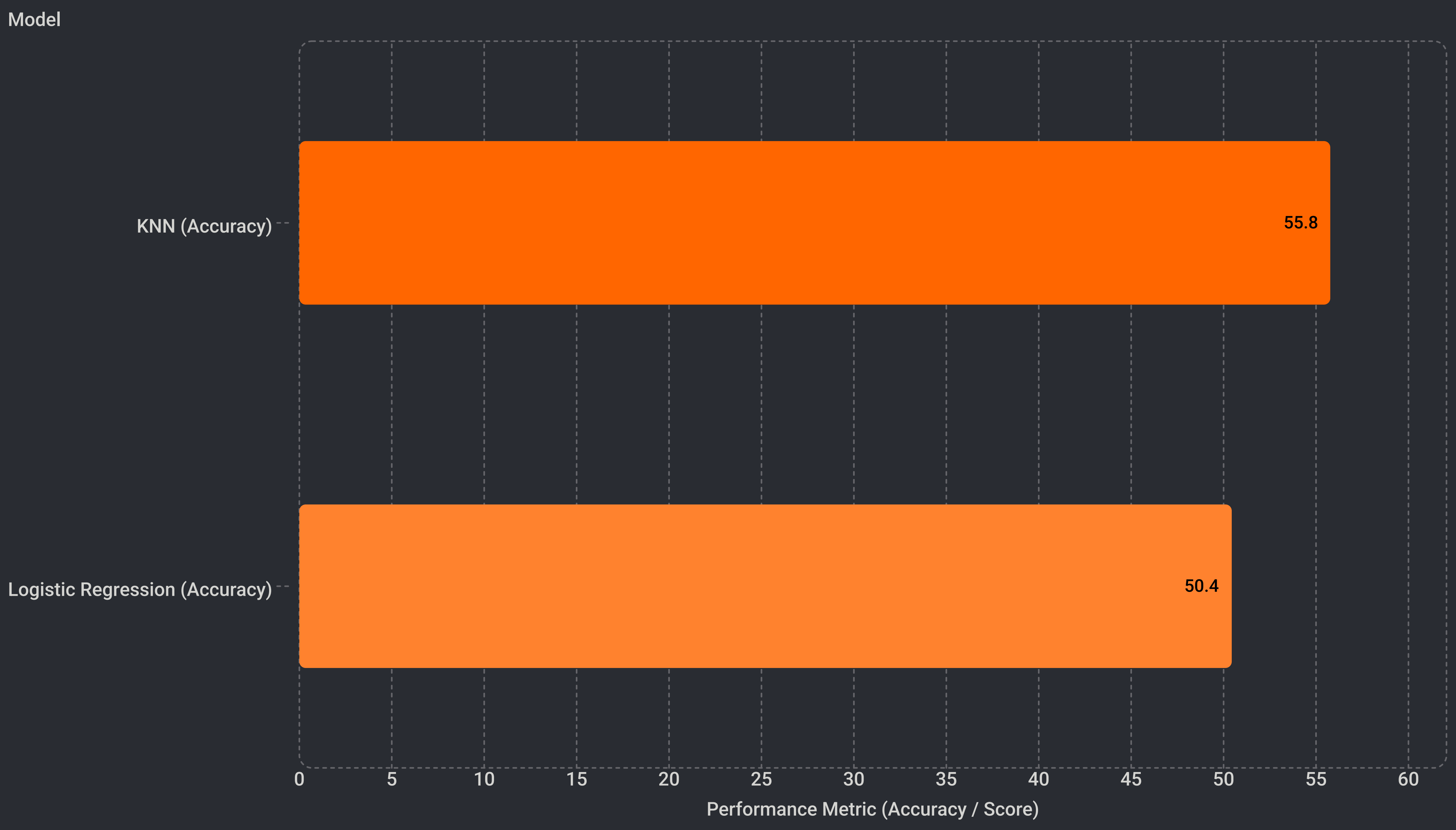
K-Nearest Neighbors (KNN)

A non-parametric classification algorithm. At k=5, it achieved an accuracy of 55.77%, making it the strongest classifier among the models tested for predicting content categories or engagement levels.

Logistic Regression

A statistical model used for binary classification. Its performance was nearly equivalent to a random guess, with an accuracy of 50.44%, indicating limited predictive power in this context.

The comparative performance of these models highlights which algorithms are most suitable for different analytical goals within social media data.



KNN outperforms Logistic Regression with an accuracy of 55.77% compared to 50.44%, indicating that KNN's non-parametric approach is better suited for classifying content categories and engagement levels in this social media dataset. While both models show modest predictive power, KNN's 5.33 percentage point advantage suggests it captures non-linear patterns in the data more effectively than the linear assumptions of Logistic Regression.

Conclusion

Objective 1 — Content/Hashtag/Platform Strategy

Video, beauty/lifestyle, morning timing are top performers

Objective 2 — Demographics/Geography

19–35 age group and global audience footprint mapped; gender not a significant driver

Objective 3 — Sponsorship ROI

Brand engagement ranked, but sponsor-category data needs cleanup

Objective 4 — Timing/Post-Level Drivers

Morning + video are key levers; numeric features alone insufficient, pointing to content/text features next

Summary

Project Summary

Analyzed **52,214 posts** from **5,000 creators** across **5 platforms**, including **22,314 sponsorship deals**.

Winning Formula

Video + morning posting + beauty/lifestyle content for 19–35 year-olds is the strongest combination.

Platform Strategy

Platform choice matters for views, so YouTube and TikTok should be selected by campaign goal rather than habit.

Engagement Insight

Follower count alone is a weak predictor of engagement.

Data & Automation

SQL automation now enforces compliance and data integrity.

Sponsorship ROI Optimization

Travel and gaming categories show strong engagement potential (72.5–73.2%) — strategic sponsorship allocation to these segments can unlock additional revenue streams and improve overall ROI.

Business Recommendations

Prioritize Video + Morning

Video content and morning posting slots are the strongest engagement drivers found across all platforms

Focus on 19–35 Age Bracket

Focus beauty and lifestyle campaigns on the 19–35 age bracket — the highest-engagement demographic

Choose Platform by Goal

YouTube vs TikTok differ significantly in views — choose platform by campaign goal, not habit

Prioritize Food Category Sponsorships

Food category leads with 76.3% engagement rate, significantly outperforming other categories. Allocate premium sponsorship budgets to food brands and creators in this space for maximum ROI.

Optimize Category-Specific Sponsorship Strategy

Electronics (74.8%) and fashion (73.5%) show strong engagement, while cosmetics, gaming, and travel (72.5–73.2%) require different promotional approaches. Tailor sponsorship terms and pricing by category performance.

Thank You

Thank you for your attention and support throughout this presentation.

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